COMPUTER NETWORKS (DU #2304CS301)

Unit-3: Network Communication Reference Models

Complete Exam-Ready Study Guide & Practice Kit

UNIT OVERVIEW & EXAM OBJECTIVES

Course Outcome 3 (CO3): At the end of this unit, students will be able to discuss the OSI layer and TCP/IP protocol models of computer networks along with their complete functionalities.

Primary Reference: Prof. Hemang R. Chath, Computer Science & Engineering Department, Darshan University.

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Chapter 1: Transmission Communication Modes

Communication Modes (also known as transmission modes) describe how data is exchanged between two devices over a communication channel. They specify whether data moves in one direction or both directions, and whether connected devices can transmit data simultaneously (at the same time). Transmission modes define the exact roles of the sender and receiver during data transmission.

1.1 Simplex Mode

In **Simplex mode**, communication is strictly **unidirectional**, meaning data flows in only one direction. A device can only send data or only receive data, but it can never do both. The sender and receiver have fixed roles throughout the communication.

• **Bandwidth Utilization:** Since data travels in only one direction, the entire bandwidth of the physical communication channel is dedicated to single-direction transmission, maximizing efficiency for that direction.

• **Key Advantage:** The full capacity of the channel is dedicated to one-way transmission.

• **Key Disadvantage:** Two-way communication is impossible; the receiver cannot send feedback or request retransmission.

• **Real-World Examples:** Keyboard to Computer (keyboard only sends data), Computer to Monitor (monitor only displays received data), TV and Radio Broadcasting (station transmits signals, home devices only receive).

1.2 Half-Duplex Mode

In **Half-Duplex mode**, communication is **bidirectional**, meaning both connected devices can send and receive data, **but not at the same time**. Only one device can transmit while the other receives. The full channel bandwidth is utilized for one direction at a time.

• **Error Detection & Retransmission:** Since both devices can send and receive, error detection and retransmission are possible. If the receiver detects corrupted data, it can send an error notification requesting retransmission once the channel becomes free.

• **Key Advantage:** Enables two-way communication using a single communication channel.

• **Key Disadvantage:** Simultaneous transmission is impossible; devices must wait for the channel to become idle.

• **Real-World Example:** Walkie-talkies and CB Radios. One user speaks ('Over') while the other listens; they cannot speak simultaneously.

1.3 Full-Duplex Mode

In **Full-Duplex mode**, communication is **bidirectional and simultaneous**. Both devices can send and receive data at the exact same time.

• **Channel Mechanics:** Full-duplex utilizes two separate physical channels (or two logical paths split within a single medium)—one reserved for transmitting data and the other reserved for receiving data.

• **Key Advantage:** Fastest and most efficient transmission mode with zero waiting delay.

• **Key Disadvantage:** More expensive due to the requirement of dual transmission paths or complex multiplexing.

• **Real-World Examples:** Telephone Calls, Mobile Networks, Video Conferencing (both parties can speak and hear at the same time).

Master Comparison of Transmission Modes

Feature	Simplex Mode	Half-Duplex Mode	Full-Duplex Mode
Direction	One-way (Unidirectional)	Two-way (Bidirectional, alternate)	Two-way (Bidirectional, simultaneous)
Simultaneous Tx	Not possible	Not possible (One at a time)	Possible (Simultaneous)
Data Flow	Sender → Receiver	Sender ↔ Receiver (Alternating)	Sender ↔ Receiver (Simultaneous)
Device Roles	Fixed (One sender, one receiver)	Devices act as sender/receiver alternately	Devices act as sender/receiver simultaneously
Channel Usage	Full channel used for 1-way flow	Full channel used for 1 direction at a time	Channel split into 2 paths (Tx / Rx)
Examples	Keyboard → Monitor, TV Broadcast	Walkie-Talkie, CB Radio	Telephone Call, Video Conference

Chapter 2: Types of Network Switching Techniques

Switching is the process of transferring data from a source device to a destination device through an interconnected network of intermediate nodes by selecting the best available path. Each device connects to a node, forming a network topology via transmission links. Computer networks rely on two primary switching techniques: **Circuit Switching** and **Packet Switching**.

2.1 Circuit Switching

Circuit Switching is a method where a **dedicated, physical communication path (circuit)** is established between the sender and receiver before data transmission begins. This dedicated path remains continuously reserved for the

duration of the session, preventing any other device from using those links.
• **Three Phases of Circuit Switching:**
1. *Connection Establishment:* Setup phase where routing switches establish a dedicated path.
2. *Data Transfer:* Continuous stream of analog or digital data is transmitted with low, constant delay.
3. *Connection Disconnect:* Teardown phase where dedicated resources are released for other users.
• **Key Characteristics:** Not a 'store-and-forward' technique. Ideal for continuous, real-time voice communication.
• **Example:** Traditional Public Switched Telephone Network (PSTN).

2.2 Packet Switching

Packet Switching is a communication method where messages are divided into smaller, discrete data blocks called **packets**. Each packet contains the actual payload along with a header containing source IP/MAC and destination IP/MAC addresses.
• **Dynamic Routing & Store-and-Forward:** Packets travel independently through the network. Routers receive each packet, store it temporarily, inspect its header, and forward it along the optimal path. Packets of the same message may take different dynamic paths.
• **Out-of-Order Arrival & Reassembly:** Packets may arrive at the destination out of sequence or with slight congestion delays. The receiving host uses packet sequence numbers to reassemble the original message accurately.
• **Example:** The Internet, Email, Web Browsing, VoIP.

Circuit Switching vs. Packet Switching Comparison Matrix

Comparison Feature	Circuit Switching	Packet Switching
Path Reservation	Dedicated physical path established before transmission	No dedicated path; packets routed dynamically
Phases Required	3 Phases: Setup, Data Transfer, Disconnect	No setup/release phases required; direct transfer
Data Format	Continuous bit stream	Discrete packets with header addressing
Transmission Method	Direct link (Not store-and-forward)	Store-and-forward mechanism at routers
Packet Routing & Order	All data follows the exact same fixed path in order	Packets travel alternate paths; reassembled at destination
Delay Characteristics	Low, uniform transmission delay after setup	Variable queueing delay due to network congestion
Bandwidth Efficiency	Inflexible (Incomplete bandwidth usage during silence)	High efficiency (Dynamic channel sharing among users)
Primary Application	Traditional Telephone Networks (Voice calls)	The Internet, Web browsing, Data networks

Chapter 3: Connection-Oriented vs. Connectionless Approaches

In network protocols, communication models are broadly classified into **Connection-Oriented** and **Connectionless** service paradigms, matching transport protocols like **TCP** and **UDP** respectively.

Feature / Parameter	Connection-Oriented Approach (e.g., TCP)	Connectionless Approach (e.g., UDP)
Connection Setup	Requires prior handshake (e.g., TCP 3-way handshake) before sending data	No prior connection setup; data sent immediately to receiver

Feature / Parameter	Connection-Oriented Approach (e.g., TCP)	Connectionless Approach (e.g., UDP)
Path Selection	All packets follow the same pre-established virtual/physical path	Each packet is treated as an independent datagram with dynamic path
Reliability & Delivery	Highly reliable; guarantees delivery, ordering, and duplicate detection	Best-effort delivery; no guarantee of delivery, order, or recovery
Packet Ordering	Strict in-order packet delivery using sequence numbers	Packets may arrive out-of-order or be duplicated
Congestion & Flow Control	Built-in flow control (sliding window) and congestion control	No flow control or congestion control mechanism
Overhead	Higher overhead due to header flags, sequence numbers, and ACKs	Low overhead; minimal header size (8 bytes for UDP vs 20+ for TCP)
Switching Linkage	Follows Circuit-Switching or Virtual Circuit concepts	Follows Datagram Packet-Switching concepts
Protocols & Applications	TCP, File Transfer (FTP), Web (HTTP/HTTPS), Email (SMTP)	UDP, DNS, Live Video Streaming, Online Gaming, Voice over IP

Chapter 4: The OSI 7-Layer Reference Model

The **Open Systems Interconnection (OSI) Reference Model** is a conceptual framework developed by the **International Organization for Standardization (ISO) in 1984**. It defines a universal standard for network communication to allow different computer systems to communicate seamlessly regardless of underlying hardware or software architecture.

• **Key Conceptual Rule:** The OSI model is a *reference model*, not a physical protocol. Each layer performs specific tasks and communicates strictly with its **adjacent layers** (directly above and below).

• **Mnemonic to Remember Layers (Bottom to Top):** *'**P**lease **D**o **N**ot **T**hrow **S**am's **P**izza **A**way'* (Physical, Data Link, Network, Transport, Session, Presentation, Application).

• **Layer Categorization:**

- **Hardware / Lower Layers (L1 - L3):** Physical, Data Link, Network (Handles physical transport & routing).
- **Heart of OSI Model (L4):** Transport Layer (Bridge between hardware and software layers).
- **Software / Upper Layers (L5 - L7):** Session, Presentation, Application (Handles user application data).

4.1 Layer 1: Physical Layer

The **Physical Layer** is the lowest layer (Layer 1) of the OSI model. It is responsible for establishing, maintaining, and terminating the physical connection between devices, transmitting raw unstructured bit streams (0s and 1s) over a physical medium.

• **Protocol Data Unit (PDU):** Bits.

• **Hardware Devices:** Hub, Repeater, Modem, Network Interface Card (NIC), Cables (Coaxial, Twisted Pair, Fiber Optic).

• **Six Core Functions of Physical Layer:**

1. *Line Configuration:* Defines connection type—Point-to-Point (dedicated link between 2 devices) or Multipoint (shared link).
2. *Signaling:* Determines the signal representation type (Electrical digital/analog signals, optical light pulses, or radio waves).
3. *Bit Synchronization:* Provides a common clock between sender and receiver so bits are sampled accurately.

4. *Data Rate Control:* Defines transmission speed measured in bits per second (bps).
5. *Physical Topologies:* Defines geometric layout of network devices (Star, Bus, Ring, Mesh topology).
6. *Transmission Modes:* Defines data flow direction (Simplex, Half-Duplex, Full-Duplex).

4.2 Layer 2: Data Link Layer

The **Data Link Layer (L2)** ensures error-free transfer of data frames between two directly connected nodes over a physical link. It handles **hop-to-hop (node-to-node) delivery** and physical hardware addressing.
• **Protocol Data Unit (PDU):** Frame.
• **Hardware Devices:** Layer 2 Switches, Bridges, Network Interface Cards (NICs).
• **Five Core Functions of Data Link Layer:**
1. *Framing:* Packages raw bit streams from Physical layer into structured data blocks called frames with headers and trailers.
2. *Physical (MAC) Addressing:* Adds 48-bit hardware MAC address of source and destination to frame headers.
3. *Flow Control:* Prevents a fast sender from overwhelming a slow receiver.
4. *Error Control:* Detects and retransmits damaged/lost frames.
- **Error Detection Techniques:** Cyclic Redundancy Check (CRC), Checksum.
- **Error Correction Techniques:** Hamming Code, Automatic Repeat Request (ARQ).
5. *Access Control:* Determines which device controls the link when multiple devices share a channel.
• **IEEE Data Link Sublayers:**
- **Logical Link Control (LLC) Sublayer:** Handles flow control, error checking, and link maintenance.
- **Media Access Control (MAC) Sublayer:** Handles MAC physical addressing and channel access protocol.

4.3 Layer 3: Network Layer

The **Network Layer (L3)** is responsible for **source host to destination host packet delivery** across different, interconnected networks (internetworking). It determines the optimal routing path.
• **Protocol Data Unit (PDU):** Packet.
• **Hardware Devices:** Routers, Layer 3 Switches.
• **Four Core Functions of Network Layer:**
1. *Internetworking:* Provides logical connections between heterogeneous, interconnected networks.
2. *Logical Addressing:* Adds logical IP addresses (IPv4 / IPv6) to identify hosts uniquely across global networks.
3. *Routing & Forwarding:* Selects the best/shortest path for packet delivery using routing protocols and routing tables.
4. *Packetizing (Encapsulation):* Encapsulates Transport layer segments into IP packets with source and destination headers.

4.4 Layer 4: Transport Layer ('Heart of OSI')

The **Transport Layer (L4)** is responsible for **end-to-end delivery of the complete message** from source process to destination process. It acts as the critical bridge between network hardware layers and software application layers.
• **Protocol Data Unit (PDU):** Segment (TCP) / Datagram (UDP).
• **Core Protocols:** TCP (Connection-oriented, reliable) and UDP (Connectionless, lightweight).
• **Five Core Functions of Transport Layer:**
1. *Service Point (Port) Addressing:* Uses port numbers to deliver data to specific software processes running on a device.
- **Well-Known Port Examples:** HTTP (Port 80), HTTPS (Port 443), FTP (Port 21), SMTP (Port 25), DNS (Port 53).
2. *Segmentation & Reassembly:* Splits large messages from Session layer into smaller segments with sequence numbers, reassembling them in exact order at destination.
3. *Connection Control:* Manages connection establishment, data transfer, and termination for TCP.

4. *Flow Control:* Performs end-to-end flow control to match sender rate with receiver capacity.
5. *Error Control:* Performs end-to-end error checking and automatic retransmission of corrupted/lost segments.

4.5 Layer 5: Session Layer

The **Session Layer (L5)** establishes, manages, synchronizes, and terminates communication sessions between software applications.

• **Four Core Functions:**

1. *Session Establishment:* Initializes connection between two applications.
2. *Dialog Control:* Manages direction of conversation (Simplex, Half-Duplex, or Full-Duplex dialog).
3. *Session Termination:* Safely closes session upon transfer completion.
4. *Synchronization (Checkpointing):* Inserts checkpoints into long data streams. If network crashes, transmission resumes from the last checkpoint rather than restarting from zero.

4.6 Layer 6: Presentation Layer ('Translation Layer')

The **Presentation Layer (L6)** ensures that data exchanged between applications is in a mutually readable format. It handles data formatting, syntax, and semantics.

• **Three Core Functions:**

1. *Translation:* Converts local device-dependent code formats to a standard common network format (e.g., ASCII to EBCDIC conversion).
2. *Encryption & Decryption:* Converts plaintext to ciphertext for transmission security and decrypts ciphertext at receiver using security keys.
3. *Compression & Decompression:* Reduces bit count to maximize transmission efficiency and decompresses data at receiver.

4.7 Layer 7: Application Layer

The **Application Layer (L7)** is the topmost OSI layer. It provides network services directly to end-user software applications.

• **Three Core Functions:**

1. *Network Services:* Provides email, file transfer, web access, and remote database queries.
2. *Protocol Handling:* Supports protocols like HTTP, HTTPS, FTP, SMTP, DNS, SNMP, POP3, IMAP, and Telnet.
3. *Resource Sharing:* Facilitates access to shared hardware resources like network printers and remote storage.

Master Summary Table of the OSI Model Layers

Layer # & Name	PDU	Addressing Type	Key Devices / Protocols	Primary Responsibility
L7: Application	Data / Message	User Application	HTTP, HTTPS, FTP, SMTP, DNS	Provides network services directly to end users
L6: Presentation	Data	Syntax / Format	ASCII/EBCDIC, SSL/TLS, JPEG	Data translation, encryption, and compression
L5: Session	Data	Session ID	NetBIOS, RPC, PPTP	Session establishment, dialog control, checkpoints
L4: Transport	Segment (TCP) Datagram (UDP)	Port Address (e.g. 80, 443)	TCP, UDP	End-to-end message delivery, segmentation, flow/error control

Layer # & Name	PDU	Addressing Type	Key Devices / Protocols	Primary Responsibility
L3: Network	Packet	Logical Address (IPv4/IPv6)	Routers, L3 Switches, IP, ICMP	Source-to-destination packet routing & internetworking
L2: Data Link	Frame	Physical Address (MAC 48-bit)	L2 Switches, Bridges, NIC, ARP	Hop-to-hop frame delivery, physical addressing, CRC
L1: Physical	Bits (0s/1s)	Physical Channel	Hub, Repeater, Modem, Cables	Raw bit transmission over physical medium

Chapter 5: The TCP/IP Protocol Model & Suite

The **TCP/IP (Transmission Control Protocol / Internet Protocol) Model** is the practical standard networking model used across the global Internet today. Developed by the **U.S. Department of Defense (DoD) in the 1960s**, it predates the OSI model.

• **Architectural Versions:**

- **Original TCP/IP Model (4 Layers):** Application, Transport, Internet, Link (Network Access).
- **Updated TCP/IP Model (5 Layers):** Application, Transport, Network, Data Link, Physical.

TCP/IP Protocol Suite Layer Mapping

TCP/IP Layer (5-Layer)	Corresponding OSI Layers	Key Protocols Included	Functional Summary
5. Application Layer	Application, Presentation, Session	HTTP, HTTPS, FTP, SMTP, POP3, DNS, SNMP	Handles application protocols, formatting, encryption, and user services
4. Transport Layer	Transport Layer	TCP, UDP	End-to-end process communication, port addressing, segmentation
3. Network / Internet	Network Layer	IP (IPv4/IPv6), ICMP, IGMP	Logical IP addressing, routing packets across networks, ICMP error reporting
2. Data Link Layer	Data Link Layer	Ethernet, ARP, MAC Protocol	Framing, physical MAC addressing, hop-to-hop transfer
1. Physical Layer	Physical Layer	Ethernet Cables, Fiber, Wi-Fi	Physical bit transmission over transmission media

Chapter 6: Comparison of OSI Model and TCP/IP Model

While both models utilize layered communication architectures, fundamental theoretical and structural differences exist between them.

Comparison Parameter	OSI Reference Model	TCP/IP Protocol Model
Full Form	Open Systems Interconnection Model	Transmission Control Protocol / Internet Protocol Model
Number of Layers	7 Layers	4 Layers (Original) / 5 Layers (Updated)
Developer / History	Developed by ISO in 1984	Developed by US Department of Defense (DoD) in 1960s

Comparison Parameter	OSI Reference Model	TCP/IP Protocol Model
Development Order	Model created first as a theoretical reference, then protocols designed	Protocols (TCP/IP) designed first, then model built around them
Session & Presentation	Has separate explicit Session (L5) and Presentation (L6) layers	Combines Session, Presentation, and Application into Application Layer
Protocol Dependence	Protocol Independent; strict separation of service, interface, protocol	Protocol Dependent; protocols are tightly integrated into the architecture
Network Layer Service	Network layer supports BOTH Connection-Oriented and Connectionless	Network layer provides ONLY Connectionless service (IP datagrams)
Transport Delivery Guarantee	Transport layer guarantees packet delivery (Connection-oriented design)	Transport layer offers option (TCP guarantees, UDP does not guarantee)
Practical Usage	Theoretical reference model; rarely implemented as-is today	Dominant practical internetworking standard used globally everywhere

Chapter 7: Solved University Question Bank & Practice Kit

Q1: Explain mode of communication with its types with suitable examples.

Answer: Communication mode (or transmission mode) describes how data flows between two devices over a network. 1) **Simplex:** Unidirectional flow (Sender → Receiver). Entire bandwidth used for 1-way flow. E.g., Keyboard to Computer, TV Broadcast. 2) **Half-Duplex:** Bidirectional alternate flow (Sender ↔ Receiver, one at a time). E.g., Walkie-talkie. 3) **Full-Duplex:** Bidirectional simultaneous flow (Sender ↔ Receiver at same time). Uses 2 physical/logical channels. E.g., Telephone call.

Q2: Describe Circuit Switching with a suitable explanation.

Answer: Circuit Switching establishes a dedicated physical path between sender and receiver before transmission begins. It consists of 3 phases: 1) Connection Establishment, 2) Data Transfer, 3) Connection Disconnect/Release. It provides constant bandwidth and low delay, but inefficient link usage during silence. Example: Public Switched Telephone Network (PSTN).

Q3: Write a short note on Packet Switching Technique.

Answer: Packet Switching breaks messages into small units called packets, each with source/destination headers. Packets are transmitted independently over dynamic routes using store-and-forward mechanisms at routers. Packets may arrive out of order and are reassembled at the destination using sequence numbers. Example: The Internet.

Q4: Compare Circuit Switching and Packet Switching.

Answer: Circuit switching establishes a dedicated reserved path prior to setup (3 phases) with low delay; packet switching routes packets dynamically over shared paths using store-and-forward. Circuit switching is ideal for voice; packet switching is highly efficient for data.

Q5: Discuss Connection-Oriented vs. Connectionless approach.

Answer: Connection-oriented service (e.g., TCP) requires a prior handshake, provides reliable in-order delivery, flow control, and error correction. Connectionless service (e.g., UDP) sends independent packets immediately without

handshake, offering high speed with best-effort delivery.

Q6: Describe the OSI Reference Model and functions of Physical and Network Layers.

Answer: Physical Layer (L1) transmits raw bits, defining line configuration, signaling, bit sync, and physical topology. Network Layer (L3) routes IP packets across networks using logical addressing (IPv4/IPv6), internetworking, and packetizing encapsulation.

Q7: Explain Transport Layer and its functions in brief.

Answer: Transport Layer (L4) provides process-to-process end-to-end message delivery. Core functions: 1) Service Point/Port Addressing (e.g., Port 80 for HTTP), 2) Segmentation & Reassembly, 3) Connection Control (TCP vs UDP), 4) Flow Control, 5) Error Control.

Q8: Compare OSI Model and TCP/IP Model.

Answer: OSI has 7 layers, developed by ISO in 1984 as a theoretical reference. TCP/IP has 4/5 layers, developed by DoD as a practical internet standard. TCP/IP merges OSI Session/Presentation into Application Layer.

Practice MCQs with Explanations

1. Which OSI layer is known as the 'Heart of OSI' model?
A) Physical Layer | B) Network Layer | C) Transport Layer | D) Application Layer
Correct Answer: C) Transport Layer
Explanation: The Transport layer (L4) bridges hardware layers (L1-L3) and software layers (L5-L7).

2. What is the Protocol Data Unit (PDU) at the Data Link Layer?
A) Bits | B) Frame | C) Packet | D) Segment
Correct Answer: B) Frame
Explanation: L1 = Bits, L2 = Frame, L3 = Packet, L4 = Segment/Datagram.

3. Which switching technique uses a store-and-forward mechanism?
A) Circuit Switching | B) Packet Switching | C) Line Switching | D) Simplex Switching
Correct Answer: B) Packet Switching
Explanation: Routers temporarily store incoming packets, inspect headers, and forward them along dynamic paths.

4. Which port number is used by HTTP (Hypertext Transfer Protocol)?
A) Port 21 | B) Port 25 | C) Port 80 | D) Port 443
Correct Answer: C) Port 80
Explanation: Port 80 is well-known for HTTP, Port 443 for HTTPS, Port 21 for FTP, and Port 25 for SMTP.

Chapter 8: Final Revision Sheet & Cheat Card

8.1 Well-Known Port Numbers Quick-Lookup Table

Protocol Name	Acronym	Port Number	Primary Purpose / Layer
Hypertext Transfer Protocol	HTTP	Port 80	Unencrypted Web Browsing (L7)

Protocol Name	Acronym	Port Number	Primary Purpose / Layer
Hypertext Transfer Protocol Secure	HTTPS	Port 443	Encrypted Secure Web Browsing (L7)
File Transfer Protocol	FTP	Port 21 (Control) / 20	File Upload/Download Transfer (L7)
Simple Mail Transfer Protocol	SMTP	Port 25	Sending Outgoing Emails (L7)
Post Office Protocol Version 3	POP3	Port 110	Retrieving Incoming Emails (L7)
Internet Message Access Protocol	IMAP	Port 143	Managing Emails directly on Server (L7)
Domain Name System	DNS	Port 53	Domain Name to IP Address Translation (L7)
Simple Network Management Protocol	SNMP	Port 161	Network Monitoring & Device Management (L7)

8.2 OSI Layer Quick-Reference Hardware & Addressing Summary

OSI Layer	PDU Unit	Addressing Method	Primary Hardware Devices
L7: Application	Data / Message	User Application Process	Gateway, Application Proxies
L6: Presentation	Data	Formatting / Encoding Syntax	Translation Gateways, Compression Engines
L5: Session	Data	Session ID / Checkpoint	Session Controllers
L4: Transport	Segment / Datagram	Service Point Port Number	Gateways, Firewalls
L3: Network	Packet	Logical IP Address (IPv4/IPv6)	Routers, Layer 3 Switches
L2: Data Link	Frame	Physical MAC Address (48-bit)	Layer 2 Switches, Bridges, NIC
L1: Physical	Bits (0s and 1s)	Physical Medium Channel	Hubs, Repeaters, Modems, Cables

EXAM SUCCESS SUMMARY

Exam Night Revision Checklist:

✓ Memorize layer order bottom-to-top using *'Please Do Not Throw Sam's Pizza Away'*.
✓ Be ready to draw the 7-layer OSI vs 5-layer TCP/IP side-by-side diagram.
✓ Know exact differences between Circuit Switching (3 phases, voice) and Packet Switching (store-and-forward, data).
✓ Memorize well-known port numbers: HTTP (80), HTTPS (443), FTP (21), SMTP (25).
✓ Remember: Physical Addressing = 48-bit MAC Address (L2); Logical Addressing = IP Address (L3); Service Point Addressing = Port Number (L4).